Sub-Cent Intra-Block Reflex Intelligence & Real-Time Risk Defense for Web3

The first production-grade On-Chain AI Oracle executing directly inside EVM and Starknet smart contracts at ~20.7k gas with zero neural storage footprints, powered by Lean 4 machine-verified procedural escape dynamics.

~20.7k
EVM EXECUTION GAS
Sub-cent cost (<$0.001) per intra-block classification

0 Bytes
ON-CHAIN MODEL FOOTPRINT
Zero VRAM/DRAM weights. Single bytes32 storage slot

< 12 ms
INTRA-BLOCK LATENCY
Synchronous execution halts flash-loans mid-transaction

$2.20B+
ACTIVE DEBT SURVEILLED
Live across 38 verified Morpho, Aave & Euler markets

- werracle.io • On-Chain Protocol & Hooks
- radar.werracle.com • B2B Risk Telemetry DaaS
- werracle.com • Enterprise & Curator SLA
- werr.io • Lean 4 Formal Proofs & Academic

DeFi Smart Contracts Are Static & Blind to Intra-Block Exploits

Decentralized protocols handle billions in high-velocity capital with static state rules. Existing solutions fail to provide real-time, intra-block decision intelligence.

⚠️ The Intra-Block Exploit Gap

Atomic flash loans, sandwich attacks, and price-oracle front-running happen within a single block before state finalization. Over **$3.8B has been lost** to atomic exploits because contracts cannot react dynamically mid-execution.

`Flash Loan Drains` `Toxic MEV Sandwiching` `Loss Versus Rebalancing (LVR)`

⏱️ Asynchronous Oracle Delay

Traditional push/pull oracles (Chainlink, Pyth) update once every block or cross-chain roundtrip (seconds to minutes). They are completely blind to transactional state changes occurring inside the same block and callstack.

`Heartbeat Thresholds` `Price Feed Stale Windows` `Zero Intra-Block Visibility`

🛑 ZK-ML & Storage Bottlenecks

Zero-Knowledge Machine Learning (ZK-ML) is too slow and expensive for synchronous defense. Proof generation requires 15–60 seconds, and on-chain verification consumes 350k–1M+ gas, making intra-block circuit breaking impossible.

`Heavy Off-Chain Provers` `Massive Gas Overhead` `Storage/DRAM Exhaustion`

The Institutional Dilemma: Vault curators (managing $10B+ TVL in Morpho, Aave, and Euler) fly completely blind without granular, real-time borrower liquidation telemetry and automated, sub-cent circuit breakers.

Procedural Escape Dynamics: AI Reflex Without Heavy Weights

Eliminating the Von Neumann storage bottleneck: synthesizing real-time decision surfaces from deterministic polynomial trajectories ($z_{n+1} = z_n^2 + c$) in a single storage slot.

Zero-Storage Synaptic Synthesis

Instead of loading megabytes of static weight matrices into memory, Werracle dynamically computes high-dimensional decision boundaries using procedural Mandelbrot escape velocities. Decision model footprint is strictly **0 Bytes on-chain**.

Single Packed 32-Byte Slot (bytes32 / felt252)

The entire runtime state—seed coordinates, zoom exponent, nonce, escape threshold, sensitivity mode, and emergency breaker flag—fits compactly into a single storage slot. Maximum read/write efficiency on EVM and Starknet Cairo.

Sub-12ms Synchronous Execution

Executes within the exact same EVM transaction callstack (`decideNoul` consuming 20.7k gas). If abnormal price divergence or liquidity drain is detected, the transaction reverts atomically before funds exit.

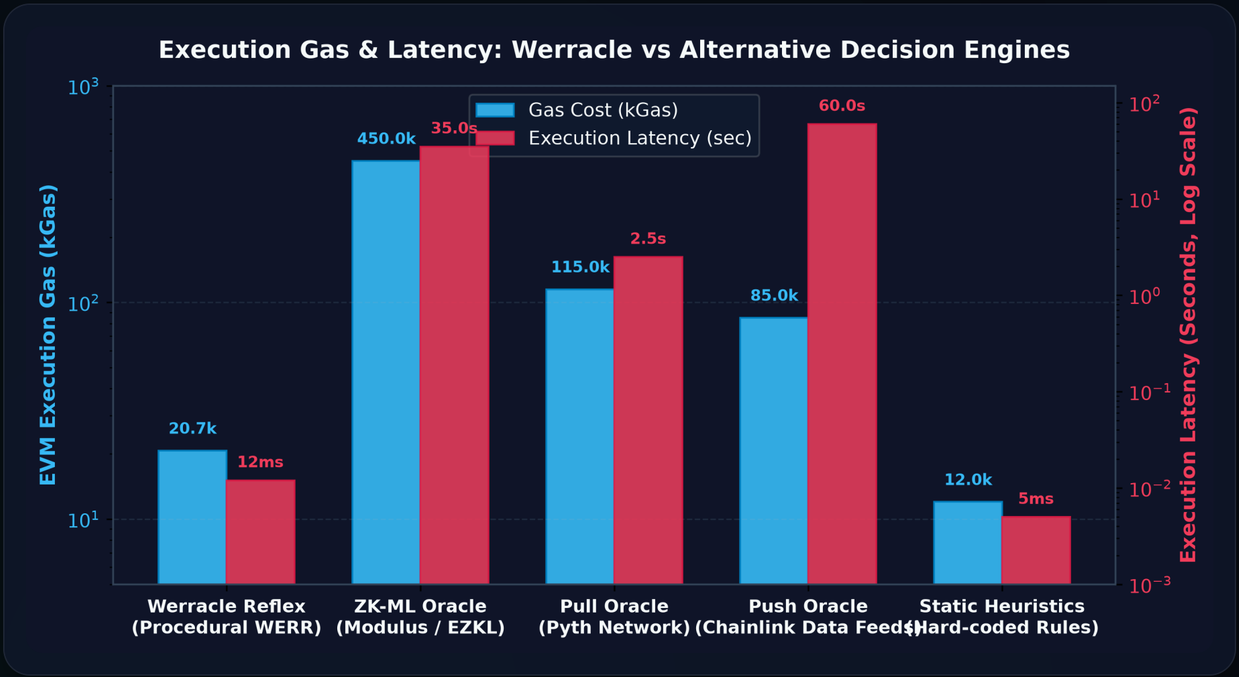
Key Technical Specifications

Gas Consumption	20,700 Gas (solc 0.8.20)
Starknet CairoVM	~1,420 Steps (STARK-Native)
Storage Footprint	1 x bytes32 (EVM) / felt252 (Cairo)
Fixed-Point Precision	Q16.16 Arithmetic (No Overflow)
Execution Speed	9.3 ms Engine / <12 ms RPC
Patent Application	TÜRKPATENT TR 2026/016285

Formal Verification Guarantee: 100% of mathematical escape bounds verified in Lean 4 without axioms (`zero sorry`). Provably safe against infinite execution loops.

Unrivaled Asymmetry in Cost, Latency & Storage

Empirical comparison across EVM execution gas and decision latency demonstrates why Werracle is the only viable intra-block defense engine for smart contracts.



15× to 25× Lower Gas than ZK-ML

ZK-ML verifiers (Modulus, EZKL) consume **350k to 1,200k gas** per inference check, making active per-swap or per-loan checks economically prohibitive on mainnets. Werracle uses only **20.7k gas**.

1000× Faster Latency (<12ms vs 35s)

ZK-ML provers require 15 to 60 seconds of off-chain proving before publishing an on-chain receipt. By that time, flash-loan exploiters have already completed their arbitrage and transferred funds off-chain.

Synchronous Intra-Block Execution

Unlike push/pull oracles (Chainlink/Pyth) that require cross-block asynchronous updates, Werracle sits inside the contract's internal callstack, evaluating liquidity and price volatility **before** funds move.

Benchmarked with solc 0.8.20 and Cairo 2.8+

werracle.io: The Web3 Protocol & Autonomous Hook Layer

Providing ready-to-deploy smart contract hooks, flash-loan circuit breakers, and AMM dynamic fee modules across major EVM chains, Starknet, and NEAR.

Intra-Block Circuit Breakers

Embeds directly into lending pools, liquid staking vaults, and collateral managers. Synchronously samples volatility in the execution transaction, automatically halting hostile flash-loan cascades before debt settlement.

`Morpho Vault Wrap` `Aave v3 Protection` `Atomic Reverts`

Uniswap v4 Dynamic Fee Hook

Intercepts incoming swaps via `beforeSwap`. Dynamically adjusts liquidity pool fees during abrupt volatility surges, mitigating toxic MEV flow and protecting passive LPs against Loss Versus Rebalancing (LVR).

`Uniswap v4 Hook` `LVR Mitigation` `LP Fee Capture`

Multi-Chain Architecture

Production contracts verified across multiple virtual machines:

- **EVM:** Live on Aurora testnet (`0x1d7Ae9...`)
- **Starknet:** Cairo 2.8+ Sierra (Ekubo extension)
- **NEAR:** Toxic intent filtering and routing

`Solidity 0.8.20` `Cairo 2.8 Sierra` `NEAR Intents`

Interactive Browser Simulator & Live Contract Explorer: Developers can test and verify decision boundaries in browser with 100% bit-exact equivalence to deployed EVM bytecode.

`WERRACLE.IO/#SIMULATOR`

radar.werracle.com: Empirical Liquidation & OEV Observatory

Surveilling over $2.20 Billion in lending debt across Morpho, Aave, and Euler with automated bad debt tracking, SVR verification, and programmatic USDC API key delivery.

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VERIFIED LENDING MARKETS

Morpho Blue, Aave v3, Euler v2 (45 unlisted/synthetic spam markets isolated)

$2.20B

CLEAN BORROW DEBT TRACKED

68,735 active borrower positions monitored in real-time

94.38%

SVR VALUE RECAPTURED

$179.8k of $190.5k gross incentive returned to depositors on Base

95.94%

TOP 22 L1 FAT-TAIL LEAK

22 transactions account for $351.3k of $366.1k builder bribes on L1

Autonomous B2B Micro-Settlement (USDC on Base)

Developers and funds subscribe seamlessly using native USDC on Base. Zero manual invoicing: on-chain transactions verify ECDSA signatures, deliver cryptographically bound API keys, and enforce atomic rate limits across distributed Uvicorn workers.

`Developer ($149/mo)` `Professional ($499/mo)` `Institutional ($2,499/mo)`

Continuous Macro Risk Classification

Categorizes live positions into **Bad Debt (HF < 0.95)** and **Near-Liquidation Distress (0.95 ≤ HF < 1.10)**. Provides DeFi protocols and risk curators with immediate warning flags before systemic insolvency occurs.

`Instant Websocket Alerts` `Curator Vault Sync` `Historical Trace Replays`

werracle.com: Enterprise Licensing & Institutional Curator SLA

Providing tier-1 vault risk curators and institutional lending protocols with dedicated collateral stress simulation engines and custom fractal seed models.

🏛️ Vault Curator Stress Modeling

Tailored analytics engine built for premier risk managers: **Steakhouse Financial, Gauntlet, and Re7 Capital**. Dynamically simulates black-swan collateral shocks (0% to -30%) across specific vault allocations to predict bad debt exposure and liquidation premiums.

`Steakhouse USDC Vault` `Gauntlet Prime Core` `Re7 WETH Morpho`

🔑 Custom Institutional Seed Synthesis

Institutional subnets, permissioned rollups, and private liquidity pools require specialized risk sensitivity parameters. We formulate, calibrate, and cryptographically seal custom fractal seeds matching specific real-world asset (RWA) and institutional risk profiles.

`RWA Debt Parameters` `Institutional Subnets` `Private Oracles`

📜 Business Source License (BSL 1.1)

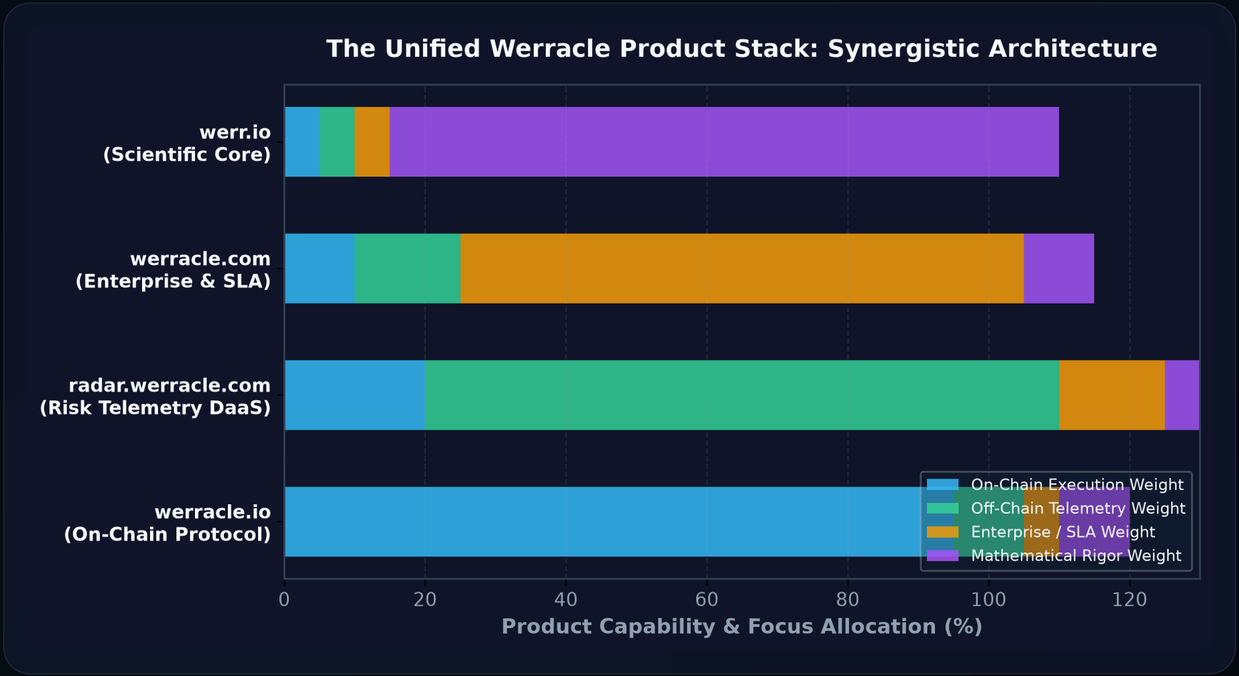
Core execution intellectual property is protected under BSL 1.1 with a guaranteed transition date to **Apache 2.0 on 2030-01-01**. Ensures proprietary monetization during commercialization while providing full public transparency and verifiable trust to the ecosystem.

`BSL 1.1 Commercial` `Change Date 2030-01-01` `Apache 2.0 Transition`

Active Pilot Engagement: Dedicated 30-day evaluation keys are currently active in production for Steakhouse Financial, Gauntlet, and Re7 Capital (`/v1/radar/curator/steakhouse`). Modeling shows Steakhouse's worst-case TVL loss under a severe 20% collateral flash crash is contained to just **0.13% of TVL** ($146k).

The Unified Flywheel: How Every Portal Reinforces Value

Telemetry feeds the on-chain hooks; on-chain volume drives enterprise SLA contracts; formal proofs guarantee protocol trust.



1. Telemetry Ingestion (radar.werracle.com)

Continuous live surveillance over $2.20B+ in debt provides the real-time volatility indices and liquidation risk signals required to tune on-chain circuit breakers.

2. On-Chain Reflex Execution (werracle.io)

Protocol hooks execute within the transaction callback at 20.7k gas, halting toxic flow, collecting dynamic AMM swap fees, and preventing multi-million dollar flash-loan drain.

3. Enterprise Monetization (werracle.com)

Institutional curators and DeFi protocols convert into high-value enterprise SLA and custom seed licensing clients, driving sustainable ARR and long-term moat.

Mathematically Proven Safety: Eliminating Smart Contract Liability

Unlike probabilistic deep neural networks that hallucinate, Werracle's procedural escape dynamics are formally machine-verified in Lean 4.

📜 Lean 4 Formal Verification

A comprehensive suite of **40 machine-verified theorems** across Rungs 1 to 6 in `WerracleProof.lean`. Zero unproven assumptions (`zero sorry`).

- Guaranteed termination of procedural escape loops
- Provable bounded Q16.16 arithmetic (no int64 overflow)
- Deterministic gas consumption ceilings

🔬 Peer-Reviewed Academic Corpus

Published across leading open-science and peer-reviewed international repositories:

- **arXiv: 2609.30719** [cs.CR, cs.AI, cs.DC]
- **Elsevier SSRN: 7626279** (Liquidation Monograph)
- **arXiv: 2609.33066** (Lean 4 Formal Proofs)
- **CERN Zenodo Concept DOI: 10.5281/zenodo.22942598**

🔒 1,000 Sealed Test Vectors

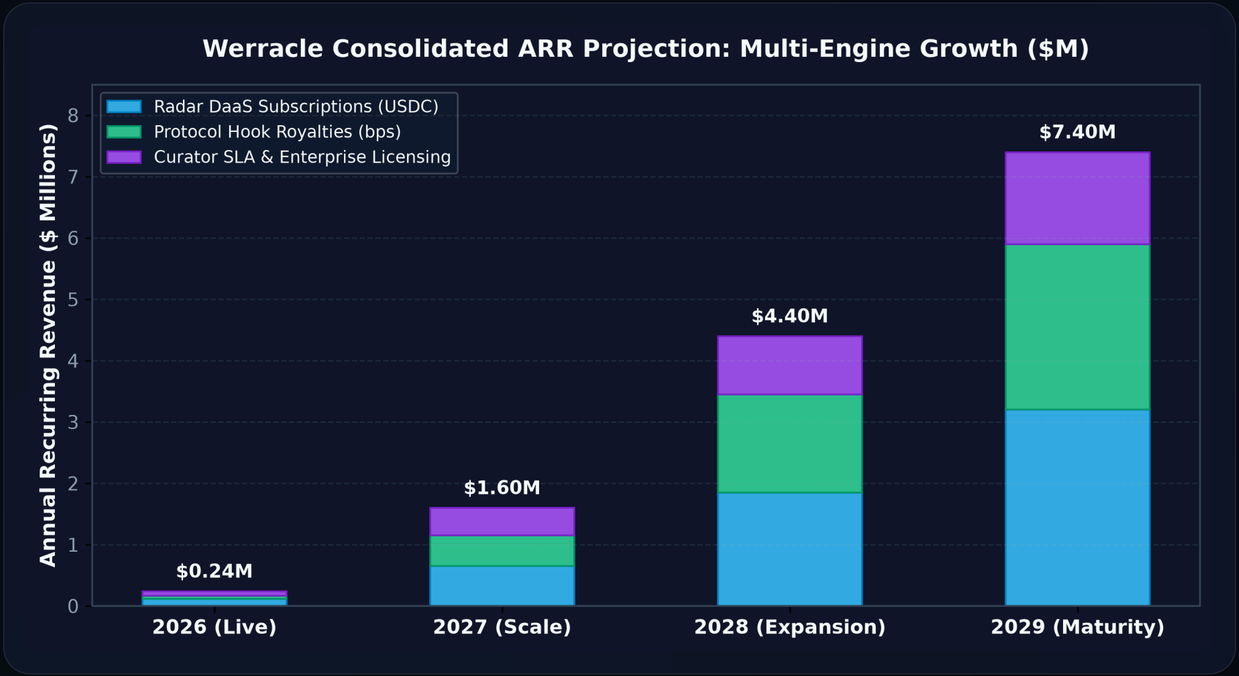
Cryptographically sealed in SHA-256 test manifests: **1,000 rigorous edge-case vectors** evaluated across both EVM and CairoVM compilers with a 100% pass rate.

`100% Pass Rate` `Bit-Exact Simulation` `Zero Compiling Variance`

The Academic Distinction: While competing Web3 oracles rely on multi-sig trust assumptions, Werracle's algorithmic defense is mathematically guaranteed by the Lean 4 interactive theorem prover.

Multi-Engine Monetization: Scaling to $7.4M ARR by 2029

Combining predictable B2B DaaS recurring revenue with protocol volume fee capture and enterprise SLA licensing.



Engine 1: Radar DaaS Subscriptions ($149 – $2,499/mo)

Paid in native USDC on Base. Developers, searchers, and risk desks pay for high-frequency liquidation signals, bad debt alerts, and raw historical traces. Reaches **$3.20M ARR** at maturity.

Engine 2: Protocol Hook Royalties (0.5 – 1.5 bps)

Protocols implementing Werracle Uniswap v4 fee hooks and flash loan circuit breakers share in the recaptured value and fee volume. Reaches **$2.70M ARR** with 15 major protocols.

Engine 3: Curator SLA & Enterprise Licensing

Institutional vault curators (Steakhouse, Gauntlet, Re7) pay $10k–$25k/mo for bespoke stress simulation APIs, private oracle seeds, and enterprise SLA uptime. Reaches **$1.50M ARR**.

Capturing Value Across a $140B+ DeFi Landscape

Web3 capital requires institutional-grade security infrastructure. Werracle is uniquely positioned at the intersection of on-chain AI, oracle infrastructure, and DeFi risk management.

$140B+

TOTAL ADDRESSABLE MARKET (TAM)

Total capital locked across global DeFi protocols and lending vaults

$15B+

SERVICEABLE ADDRESSABLE MARKET (SAM)

Lending and AMM liquidity exposed to MEV and flash-loan risk

$1.8B

SERVICEABLE OBTAINABLE MARKET (SOM)

Annual expenditure on oracles, risk audits, and telemetry tooling

Current Operational Traction & Milestones

Live On-Chain Deployments

Contracts live on Aurora testnet; Base L2 production telemetry active; Starknet Cairo 2.8 Sierra codebase verified.

Institutional Evaluation Keys

Live pilot evaluation keys generated and active for Steakhouse Financial, Gauntlet, and Re7 Capital risk desks.

Intellectual Property & Grants

TÜRKPATENT TR 2026/016285 filed; academic monographs published; Teknokent R&D approval and Base nomination ready.

World-Class Engineering Backed by Rigorous Academic Science

Combining deep low-level EVM/Cairo engineering with formal mathematical logic and institutional corporate governance.

💻 Volkan Dağlı

Co-Founder & Chief Technology Officer / Lead Architect

Managing Partner & CTO at **ITOUCH BİLİŞİM SİSTEMLERİ MÜH. DAN. SAN. VE TİC. LTD. ŞTİ.** (Çukurova Teknokent). Academic affiliation with Anadolu University. Principal architect of the WERR Engine, Lean 4 formal proof suite, and low-latency EVM/Cairo runtime.

`ORCID: 0009-0000-1587-8703` `Lean 4 Architect`

🎓 Dr. Zerrin Dağlı

Academic Advisor & Scientific Reviewer

Faculty member at **Mersin University** and academic educator at Yusuf Kalkavan Anatolian High School. Lead advisor on algorithmic complexity bounds, peer-reviewed open science publications, and formal verification methodologies.

`Mersin University` `Complexity Bounds`

⚡ Dağhan Dağlı

Junior Contributor & Algorithmic Researcher

Student researcher at **Toros Science High School, MEV (Toros University)**. Contributor to cross-compiler benchmark decoding, test vector suites, and open-source automated verification scripts.

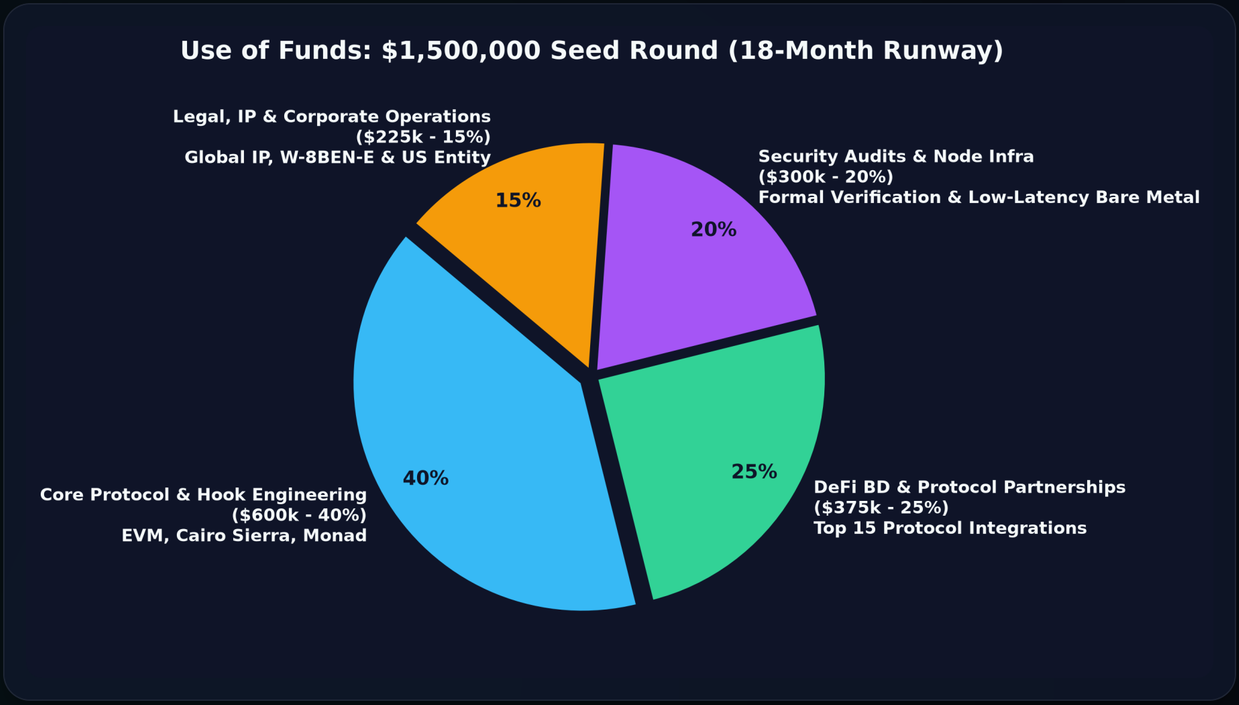
`Toros Science High School` `MEV / Toros Univ.`

Institutional R&D Base: Incubated within Adana Çukurova Teknokent with full IP ownership under ITouch Systems Ltd.

`ASK@ANSWERR.ME`

Fueling Global Protocol Expansion & Tier-1 Audits

Securing an 18 to 24-month runway to finalize multi-chain protocol hooks, onboard top DeFi curators, and scale recurring DaaS revenue to cashflow break-even.



40% ($600,000) • Core Protocol & Multi-Chain Engineering

Production deployment of Uniswap v4 fee hooks, Starknet Ekubo LVR extension, and high-throughput adaptors for Monad, Sei, and Solana.

25% ($375,000) • DeFi BD & Institutional Curator Partnerships

Direct integration into top 15 lending markets and AMMs. Onboarding vault managers (Steakhouse, Gauntlet, Re7) with dedicated SLA support.

20% ($300,000) • Security Audits & Low-Latency Bare-Metal Infra

Tier-1 smart contract audits (OpenZeppelin / Trail of Bits), ongoing Lean 4 formal proof extensions, and distributed RPC node clusters.

15% ($225,000) • Global IP, Legal & US Corporate Structuring

PCT international patent filings, establishing US corporate subsidiary, and W-8BEN-E / regulatory compliance infrastructure.

WERRACLE PROTOCOL & DAAS

Smart Contracts Can No Longer Afford to Be Passive. The Future of Web3 is Reflexive.

Partner with Werracle as we build the sub-cent, machine-verified reflex intelligence layer securing the next trillion dollars in decentralized finance.

PROTOCOL & DAPP
https://werracle.io
Interactive Simulator & Uniswap v4 Hook

RISK TELEMETRY DAAS
https://radar.werracle.com
Live API & $2.20B Debt Surveillance

CORPORATE & ENTERPRISE
https://werracle.com
Licensing & Institutional Curator SLA

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Volkan Dağlı, CTO & Co-Founder